# IFA BORU
## SPECIAL BOARDING SECONDARY SCHOOL

**Static and Current Electricity**
**Worksheet**
**Prepared For Summer Program**

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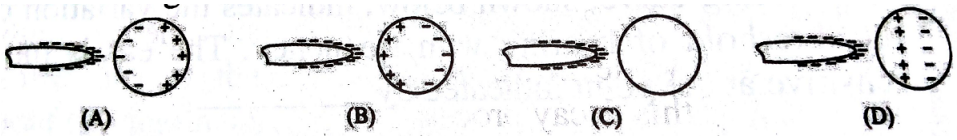
# Electric Charge, Force, Field, and Flux

1. An electron enters a region of uniform electric field of $5x10^3$ N/C. what is the force on the electron?
   A) $8x10^{-22}$ N  B) $5.3x10^{-22}$ N  C) $8x10^{-16}$ N  D) $1.88x10^{22}$ N

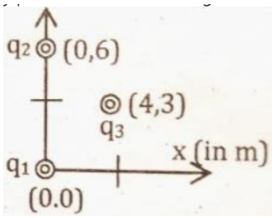
2. The distance between two point charges is doubled. What will happen to the force of interaction between the charges?

   A) The force will be doubled.
   B) The force will be reduced by half.
   C) The force will be reduced by a factor of $\frac{1}{4}$.
   D) The force will not be affected.

3. In the sketch below, which best describe the location of the negative charges (electrons) when negatively charged rod id brought near but not touching the conducting sphere?




4. A charge of $q_1 = -10.0x10^{-9}C$, $q_2 = +10.0x10^{-9}C$ and $q_3 = -2.5x10^{-8}C$ are placed on x-y plane as shown in the figure below. What is the electric force on $q_3$ by $q_1$ and $q_2$?

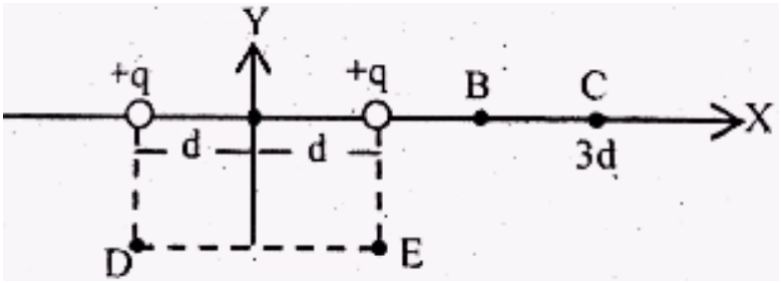



   A) $1.08x10^{-7}$ N in +ve y direction
   B) $9.00x10^{-7}$ N in -ve y direction
   C) $1.42x10^{-7}$ N in +ve x direction
   D) $1.42x10^{-7}$ N in -ve x direction

5. Suppose that three point charges are placed along a straight line and electrostatic force exerted on the middle charge is zero. What can we conclude about the charges?

   A) The signs of the three charges should be the same
   B) The charges at the ends must have the same sign
   C) The middle charge should be placed at the center between the two charges
   D) The charge at the middle and at the left end should have the same sign

6. Two positive charges of magnitude q are each a distance d from the origin A of a coordinate system as shown below. The magnitude of the electric field is minimum at point:




A) A B) B C) C D) D

7. Which one of the following statement is correct?

   A) The electric field strength due to two point charges of equal magnitude and the same sign is zero only at one point on the line joining the two charges.

   A) If a point charge of mass m is released from the rest in uniform electric field, it will move along a line of force with uniform speed.

   C) The variation of the strength of the electric field due to an electric dipole is inversely proportional with the square of the distance from its center of mass.

   D) An electric dipole moment p in a uniform external electric field E is in stable equilibrium if p and E are in opposite directions.

8. A positively charged glass rod and a negatively charged rubber rod are placed far from each other. What happens to the dry pieces of paper brought nearby the rods before touching them?

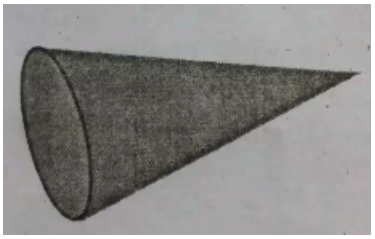
   A) repelled by glass and attracted by rubber
   B) attracted by glass and repelled by rubber
   C) attracted by both glass and rubber
   D) repelled by both glass and rubber

9. Two particles, with charges $q_1$ and $q_2$, are on the x-axis, with $q_1$ at $x = a$ and $q_2$ at $x = -2a$. For net force on a third charged particle, at x=0, to be zero, q1 and q2 must be related by:

   A) $q_2 = 2q_1$ B) $q_2 = -4q_1$ C) $q_2 = -2q_1$ D) $q_2 = 4q_1$

10. A point charge $Q_1$ is at x = 0 and $Q_2$ is at x = 8cm. What is the relationship between these charges if the resultant electric field strength is zero at x = 16cm?
    A) $Q_1 = 2Q_2$ B) $Q_1 = -0.25Q_2$ C) $Q_2 = 0.25Q_1$ D) $Q_1 = -4Q_2$

11. The cone shaped solid conductor shown below is negatively charged. Which of the following statement is correct?



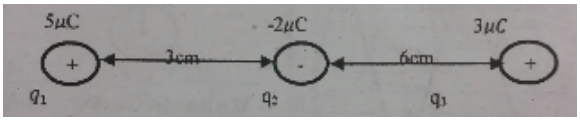
   A) the charge density is greater at the tip of the conductor.

B) the charge is uniformly distributed on its surface.

C) the electric field is very weak at the tip of the conductor.

D) there is an electric field inside the conductor.

12. Three point charge lie in a straight line as shown in the figure below. What is the net force acting on the $-2\mu C$ charge? (Use $K = 9x109Nm^2/C^2$)




A) 100N, to the left B) 15N, to the right
C) 115N, to the right D) 85N, to the left

13. Which of the following is Not true about the effect of electrostatic charges? They

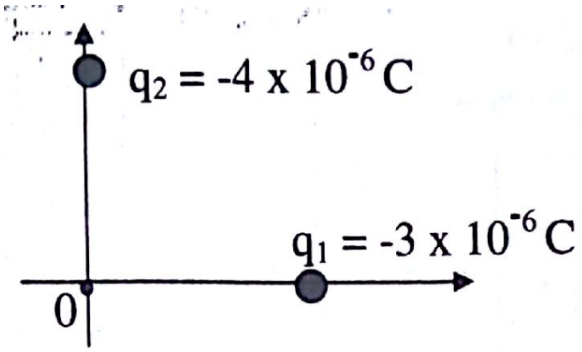
A) can be applied in the photo copying process

B) can be applied for spray painting.

C) can result in an electric current between object which is always useful and harmless.

D) can be applied to filter the chimney by trapping soot particles so that they will not pollute the air

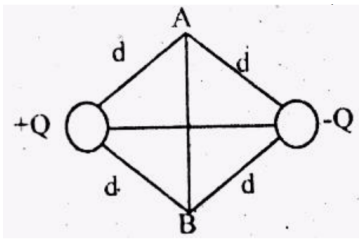
14. Charges $q_1$ and $q_2$ are located at positions (3m,0) on the x-axis and (0,3m) on the y-axis respectively as shown in the following figure. What should be the x and y components of the net electric force on a 1æc charge placed at the origin? ($k = 1/(4\pi\varepsilon_0) = 9 \times 10^9 Nm^2C^{-2}$)




A) $F_x = 4 \times 10^{-3}N$ and $F_y = 3 \times 10^{-3}N$
B) $F_x = 3 \times 10^{-3}N$ and $F_y = 4 \times 10^{-3}N$
C) $F_x = -4 \times 10^{-3}N$ and $F_y = -3 \times 10^{-3}N$
D) $F_x = -3 \times 10^{-3}N$ and $F_y = -4 \times 10^{-3}N$

# Electric Potential and Potential Energy

15. Points A and B are each a distance d from each of two equal and opposite charge as shown below. What will be the work required to move a negative charge q from A to B?




A) Zero B) $\frac{qQ}{4\pi\varepsilon_0 d^2}$ C) $\frac{qQ}{4\pi\varepsilon_0 d}$ D) $\frac{qQ}{4\pi\varepsilon_0(4d^2)}$

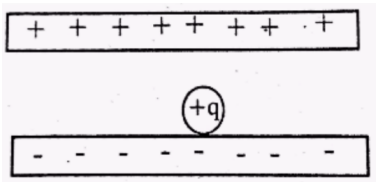
16. Moving $2.5x10^{-6}$ coulomb of charge from point A to point B in an electric field requires $6.3x10^{-4}$ joule of work. The potential difference between points A and B is approximately:

A) $4.0x10^{3}$ V B) $2.5x10^{-2}$ V C) $1.6x10^{-19}$ V D) $1.0x10^{-14}$ V

17. What is the potential (V) at a distance of 10m from a charge of +5.0C?

A) $-4.45x10^{9}$ V B) $-0.5$ V C) $+4.45x10^{9}$V D) $+0.5$V

18. Positive charge of $1.0x10^{-4}$ C is on the negatively charged plate of a parallel plate capacitor as shown below. If the potential difference on the top plate is +10,000V and that of the bottom plate is -10,000V, how much work (W) is required to move the charge from the bottom to the top plate?

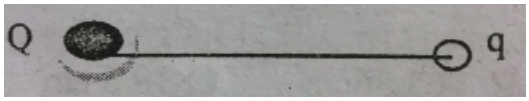



A) 2.0J B) 1.0J C) zero D) $2.0x10^{4}$J

19. Two positive charges of magnitude $2.4x10^{-9}$C and $1.6x10^{-9}$C are separated in air by a distance of 2.0cm. How much electrostatic potential is produced by the two charges at midway between them?

A) 3.6kV B) 0.72kV C) 2.16kV D) 1.44kV

20. In the figure below Q is a negative source charge and q is a positive test charge. If q is replaced by a charge of the same magnitude but negative, the potential at this position due to Q




A) remains the same B) increases C) decreases D) changes its sign

21. Which one of the following is true about absolute potential and potential difference?

A) Both quantities have arbitrary value that can be set equal to zero convenience.

B) potential difference is a location dependent quantity.

C) absolute potential is energy while potential difference is the work done per unit time.

D) both quantities are measured in the same unit even if they are different.

22. What is an electric potential energy? It is the energy a charge possesses because, it is in the
A) region of other masses
B) region of zero electric field
C) vacuum
D) region of other charges

23. Which of the following is true about an absolute electric potential? It is

A) an electric potential at a position in an electric field relative to that at infinity from the field source.

B) An electric potential at a position in an electric field relative to that at another positon in the field.

C) The work done to move a unit charge from an infinity to a position in an electric field.

D) The work done per unit charge to move it from one point to another in an electric field.

# Capacitor, Capacitance, and Dielectrics

24. An air-filled parallel-plate capacitor has a capacitance of 1pF. The plate separation is then doubled and a wax dielectric is inserted, completely filling the space between the plates. As a result, the capacitance becomes 2pF. What is the dielectric constant of the wax?

A) 8.0 B) 4.0 C) 2.0 D) 0.5

25. What is the capacitance of a capacitor that should be connected with 2 other capacitors having capacitance of $4\mu$F and $8\mu$F all in series to produce an effective capacitance of $\frac{24}{11}\mu$F?

A) 12 $\mu$F B) 10 $\mu$F C) 8 $\mu$F D) 4 $\mu$F

26. Which one of the following changes to parallel plate capacitor would not increase energy stored in the capacitor at a fixed voltage across the plates?

A) increasing the area of the plates

B) increasing the dielectric constant

C) decreasing the charges on the plates

D) increasing the distance between the plates

27. A parallel plate capacitor has a capacitance of $10\mu$F. If the distance between the plates is doubled, keeping all other quantities constant, what is the new capacitance of the capacitor?

A) 0.2 $\mu$F B) 5 $\mu$F C) 10 $\mu$F D) 20 $\mu$F

28. A capacitor of 12 $\mu$F capacitance can fully charged after it is connected between a potential difference of 9 V. how much external energy is used in charging the capacitor?

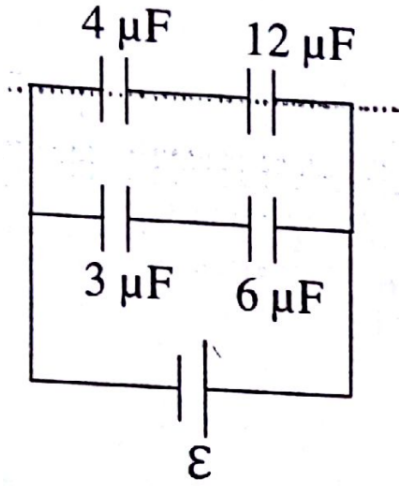
A) $1.0x10^{-3}$ J B. $54x10^{-4}$ J C. $9.72x10^{-4}$ J D. $4.86x10^{-4}$ J

29. A $2\mu$F and a $1\mu$F capacitors are connected in parallel and a potential difference is applied across the combination. The $2\mu$F capacitor has
A) half the charge of the $1\mu$F capacitor
B) twice the potential difference of the $1\mu$F capacitor
C) half the stored energy of the $1\mu$F capacitor
D) twice the stored energy of the $1\mu$F capacitor.

30. A 9V battery is connected to a $2\mu F$ capacitor. How much electric energy can be stored in the capacitor?
A) $8.1x10^{-5}J$ B) $8.1x10^{-6}J$
C) $1.62x10^{-4}J$ D) $1.62x10^{-5}J$

31. If 9V battery is connected to the plates of a $3\mu$F capacitor, what will be the charge stored in the capacitor
A) 54 $\mu$F B) 1/3 $\mu$F C) 3 $\mu$F D) 27 $\mu$F

32. The following circuit diagram shows a combination of different capacitors and a voltage sources. What is the equivalent capacitance of the circuit?

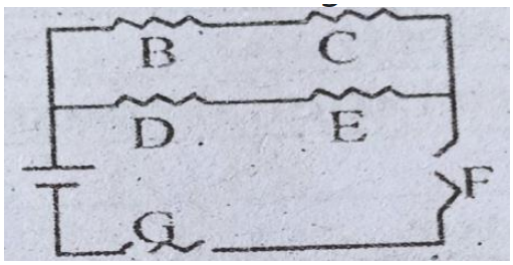



A) $16.00\mu F$ B) $27.00\mu F$ C) $5.00\mu F$ D)$5.76\mu F$

## Electric Current, Resistance, and Ohm's Law

33. Suppose that a voltage source produces 240V for a moment. By what percentage does the power output of 220V light bulb increases?
A) 38% B) 36% C) 19% D) 18%

34. If all the six resistors in the circuit shown in the figure have equal resistance, and the current flowing into resistor B is 4A, then what is the current flowing into resistor G?




A) 16A B) 8A C) 2A D) 1A

35. A small radio receiver uses a battery that delivers a constant current of 25mA for 3 hours. Calculate the total charge delivered by the battery.

A) 1080000C B) 270000C C) 540C D) 270C

36. What is the resistance of a copper cable if it has a cross sectional area of $1cm^2$ and a length of 2km? The resistivity of copper is $2x10^{-8}\Omega m$.

A) 0.0004Ω B) 0.004Ω C) 0.04Ω D) 0.4Ω

37. If the terminals of a battery with zero internal resistance are connected across two identical resistors in series, the total power delivered by the battery is 8 W. If the same battery is connected across the same resistors in parallel, what is the total power delivered by the battery?

A) 2W B) 4W C) 16W D) 32W

38. Wire B has twice the length and twice the radius wire A. Both wires are made from the same material. If wire A has a resistance R, what is the resistance of wire B?

A) 4R B) 2R C) $\frac{R}{2}$ D) $\frac{R}{4}$

39. If several resistors are connected in parallel, the equivalent resistance is

A) greater than any of the resistances in the group.
B) equal to the average resistances of the group.
C) less than any of the resistances in the group.
D) equal to the sum of the resistances in the group.

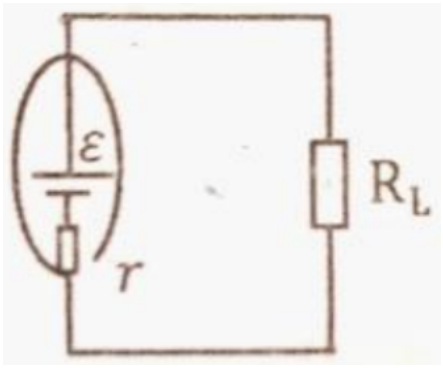
40. In aluminium and copper wires with circular cross sections have equal length and equal resistance, then the ratio of the radius of aluminium to that of copper should be the same as that of the _____

A) ratio of the resistivity of aluminium to copper
B) ratio of the resistivity of copper to aluminium
C) square root of the ratio of the resistivity of aluminium to copper
D) square root of the ratio of the resistivity of copper to aluminium

41. Three resistors connected in series have individual voltage labeled $V_1$, $V_2$ and $V_3$, respectively. Which of the following express the value of the total voltage $V_T$ taken over the three resistors together?

A) $V_T = V_1 + V_2 + V_3$ C) $V_T = V_1 = V_2 = V_3$
B) $V_T = 1/V_1 + 1/V_2 + 1/V_3$ D) $V_T = (1/V_1 + 1/V_2 + 1/V_3)^{-1}$

42. Which of the following is NOT correct about the circuit shown below?




A) Voltage across $R_L$ is $\frac{R_L}{(r+R_L)\varepsilon}$ C) Current flowing through $R_L$ is $\frac{\varepsilon}{(r+R_L)}$

B) Current flowing through r is $\frac{r}{(r+R_L)\varepsilon}$ D) Power drop in $R_L$ is $\frac{R_L}{(r+R_L)\varepsilon^2}$

43. If a cell supplies a current of $0.9A$ when connected to a $2\Omega$ resister and a current of $0.3A$, then what will be the internal resistance of the cell when connected to a $7\Omega$ resistor?

A) 2 Ω B) 12 Ω C) 1 Ω D) 0.5 Ω

44. A copper wire of cross sectional area $2mm^2$ carrying a current of 1.5 A. if the drift velocity of the electron is 0.00028 m/s, what is the number of charge carriers (free electrons) per unit volume in the copper wire?

A) $1.5x10^{25}/m^3$ B) $1.5x10^{32}/m^3$ C) $1.88x10^{28}/m^3$ D) $1.67x10^{32}/m^3$

45. A simple circuit consisted of a resistor of $10\Omega$ connect to a battery of emf 18V. if the current through the circuit is 1.6A, what is the internal resistance of the battery?

A) 10.0 Ω B) 8.75 Ω C) 1.25 Ω D) 11.2 Ω

46. Ethiopian Electric Power Corporation calculates the first 50 kwh consumption at the minimum rate. In this quantity the unit kwh refers to

A) power B) current C) charge D) energy

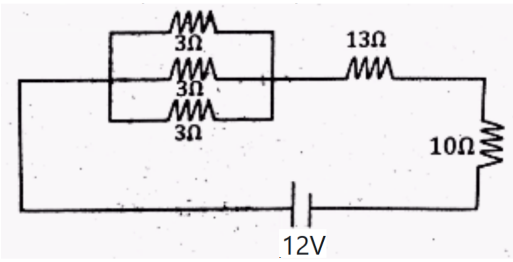
47. The terminal potential difference of ta battery is the maximum when

A) the e.m.f of the battery is the minimum.
B) the battery is being discharged through a wire.
C) there is a steady current flowing through the battery.
D) there is no current flowing through the battery.

48. A stove that draws 10A current from 220V line is used for 5 hours. If the electrical company demands 20 cents per kilowatt-hour, what is the cost of the electrical energy consumed?

A) 2 Birr and 20 cents B) 1 Birr and 10 cents
C) 1 Birr D) 55 cents

49. A 12V battery is connected to a resistors networks as shown in the figure below.

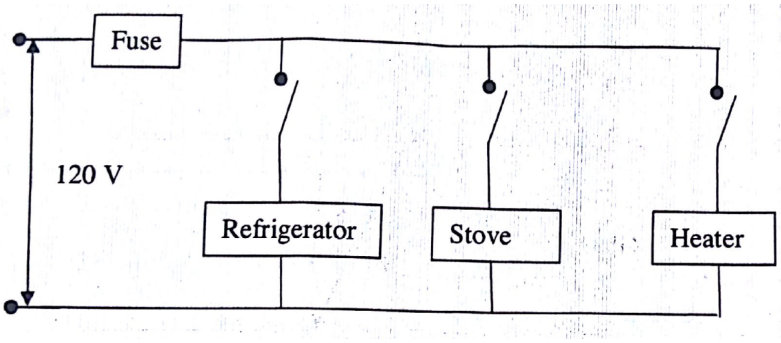


What is the potential drop across the 10Ω resistor?
A) 5V B) 1V C) 10V D) 12V

50. Which of the following statement is correct about resistivity and conductivity?
A) they are dimensionless.
B) they have the same SI unit.
C) they have direct relationship.
D) one is the reciprocal of the other.

51. The potential difference between the terminals of battery when the battery is isolated electrically is called
A) electric force B) electromotive force
C) terminal force D) electrical field

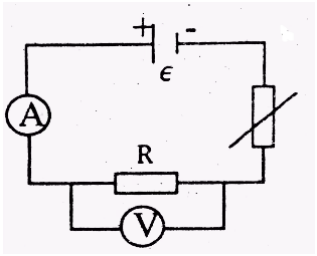
52. A 360W refrigerator, a 1200W stove, a 720w heater and a 15 A fuse are connected to 120V main line as shown in the figure. Under what condition will the fuse melt?




A) when the refrigerator and the stove are switched on.
B) when the refrigerator and the heater are switched on.
C) when the heater and the stove are switched on.
D) when any of devices is switched on.

53. The two ends of a 4Ω resistor are connected to a 16V battery. What is the total power delivered by the battery to the circuit?
A) 4W B) 16W C) 32W D) 64W

54. A group of grade ten students connected a battery, voltmeter, resistor, rheostat and ammeter in a circuit to demonstrate Ohm's law as shown in the figure below What are the data to be taken? The values of



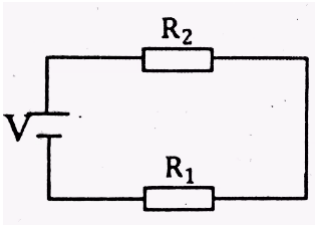

A) potential difference across R and that of rheostat.

B) Both the current and the potential difference across R for different values of rheostat.

C) The current in the circuit and the potential difference across the rheostat

D) current in the circuit and that of rheostat.

55. Two resistors and a voltage source are connected in series as shown in the figure below. If $R_1 = 2R_2$ which of the following statements is true?




A) The current in $R_1$ is half of that in $R_2$

B) The current in $R_1$ is twice of that in $R_2$

C) The potential difference across $R_1$ is half of the across $R_2$

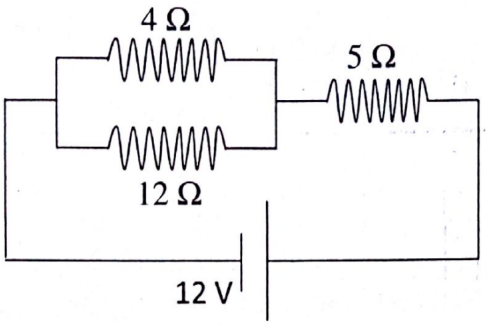
D) the potential difference across $R_1$ is twice of that across $R_2$

56. If a battery with an emf of 24v is connected to a 6Ω resistor and as a result, a current of 3A exists in the resistor, what is the internal resistance of the battery?
A) 8 Ω B) 14 Ω C) 2 Ω D) 6 Ω

57. One kilo-watt-hour (1KWh) is the same as,
A) $3.6 \times 10^3 J$ B) $3.6 \times 10^6 J$ C) $1.0J$ D) $3.6J$

58. A uniform cylindrical wire of $1cm$ radius and $10m$ length has a resistance of $5 \times 10^{-4}\Omega$, what is the resistivity of the wire? A) $1.00 \times 10^{-4}\Omega m$ B) $1.57 \times 10^{-4}\Omega m$
C) $1.00 \times 10^{-8}\Omega m$ D) $1.57 \times 10^{-8}\Omega m$

59. The circuit diagram shown below is a combination of resistors and a voltage source. The voltage drop across the 5Ω resistor is




A) 4.5V B) 2.0V C)7.5V D) 6.0V

60. A dry cell's voltage is labelled to be 1.5V. Which of the following statements is correct?

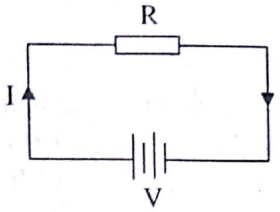
A) The terminal voltage of the cell is equal to 1.5V for closed circuit.

B) The electromotive force (emf) of the cell is equal to 1.5V.

C) The terminal voltage of the cell is greater than 1.5V.

D) The emf of the dry cell is less than 1.5V.

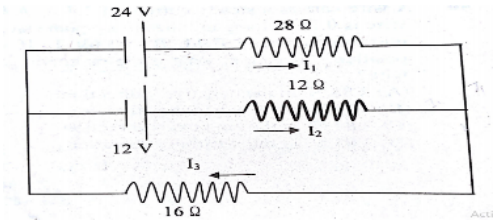
61. A resistor of resistance R is connected to a battery as shown in the figure below. If the potential difference and the current are V and I, respectively, what is the electrical power P delivered to the circuit?




A) $P = R \times V^2$ B) $P = I \times V$ C) $P = I^2/R$ D) $P = (I \times V)/R$

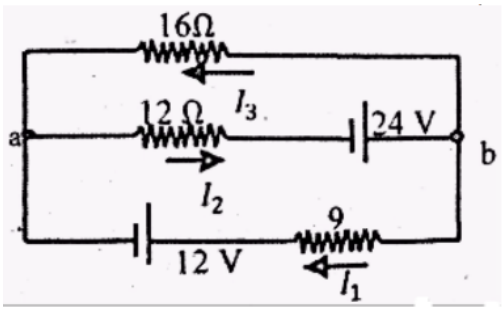
## Measuring Instruments and Kirchhoff's Rules

62. In Figure given below, the current in the 16Ω resistor is




A) 0.64A B) 0.15A C) 0.49A D) 1.28A

63. In the circuit shown below, what is the potential difference between point b and a?




A) 19.2V B) 24.0V C) 13.0V D) 12.0V

64. Which one of the following statement is not correct about galvanometer?

A) It is used to measure small amount of current.
B) It has infinite resistance
C) It can be used to measure voltage if shunted with large load in series.
D) Its working principle is the same as that of electric motor.

65. Which instrument should, ideally, have zero resistance?

A) Wheatstone bridge C) Voltmeter
B) Oscilloscope D) Ammeter

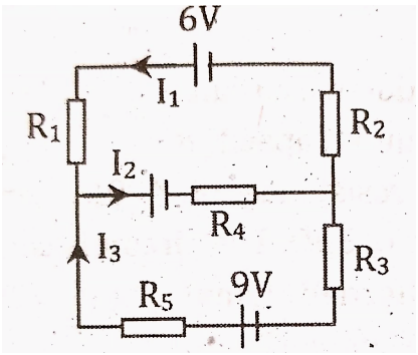
66. A galvanometer of full-scale deflection 10mA is to be converted into a 0-10.0A ammeter. If its coil has a resistance of 100Ω, then what value of shunt resistance must be fitted?

A) 0.1 Ω B) 10 Ω C) 9.99 Ω D) 0.999 Ω

67. Which of the following statement is correct?

A) Galvanometer can be converted to voltmeter of different scale by connecting low valued resistance in parallel
B) Low valued internal resistance increases accuracy of voltmeter
C) High valued internal resistance increases accuracy of ammeter
D) Galvanometer can be converted to ammeter of different scale by connecting variety of low value shunt resistance in parallel.

68. In the circuit shown below, each resistor has a value of 3Ω. What is the value of the current through $R_4$?




A) 0.75 A B) 1.0 A C) 0.625 A D) 0.125 A

69. A voltmeter should be connected in parallel across a resistor in a circuit because
A) its resistance is very small
B) it requires high current
C) its resistance is very high
D) it measures current